

Product Architecture Specification: AI Visual Co-Presenter & Visually-Grounded Summaries

Document Version: 1.0.0

Target Platform: Tauri v2 + Rust Core, Next.js / TypeScript / Tailwind CSS

Execution Environment: Local-First (Whisper.cpp, Apple Vision / Tesseract, Ollama)

1. Executive Overview

This specification defines two tightly coupled features designed to revolutionize live remote presentations and post-meeting documentation:

- The Real-Time Visual Co-Presenter:** An ambient, dual-window desktop companion that listens to spoken dialogue, detects visual intent and prosodic pauses, and surfaces verified, presentation-grade images (diagrams, schematics, clean physical cutouts) for instant single-keystroke projection to an audience.
 - Visually-Grounded Executive Summaries:** A post-meeting synthesis pipeline that leverages interleaved on-screen OCR and stage projection logs to resolve visual deictic references into dense, self-contained plain text, accompanied by a non-intrusive split-pane visual reel strictly capped at one anchor image per point.
-

2. Feature 1: Real-Time Visual Co-Presenter

2.1 The Dual-Window Presentation Model

To eliminate the clutter, latency, and privacy risks of live tab-switching and window dragging, the UI is split across two decoupled Tauri windows:

- Audience Stage Window (/stage):**
 - Target for screen-sharing software (Zoom, Google Meet, Microsoft Teams).
 - Frameless, borderless, zero UI chrome.
 - Renders visual assets centered inside an adaptive canvas container (subtle dark slate #0f172a, rounded corners, ambient drop-shadow) with smooth crossfades.
- Presenter Cockpit Window (/cockpit):**

- Displayed on the presenter's private screen.
- Displays a linear, horizontal reel:
- **Left (History Reel):** Dimmed (40%) stack of past projected visuals for instant backwards navigation (`Left Arrow`).
- **Center (Active Card):** Exactly what is currently live on the Audience Stage.
- **Right (Next Up / Filmstrip):** Incoming suggestions (Ghost or Elevated state).

2.2 Intent Detection & Linguistic Grounding

The pipeline avoids naive keyword matching and broad conversation history to prevent displaying stale context.

1. Immediate Focus Head Binding:

- A rolling 3–5 token window captures the syntactic head noun immediately adjacent to the trigger.
- *Forward-pointing (Cataphoric):* "Look at [this split keyboard layout]."
- *Backward-pointing (Anaphoric):* "...so we switched to [a cycloidal drive], like this."

2. The Two-Tier Elevation Model:

- **Tier 1: Ambient Ghost State:** During natural, uninterrupted speech, incoming concrete nouns populate the candidate slot at 30% opacity with zero intrusive audio/visual chimes.
- **Tier 2: Elevated / Focused State:** The card smoothly escalates to 100% opacity with a subtle border glow when:
 - An explicit linguistic anchor is detected ("look at this", "like a...", "here is the...").
 - **Prosodic Pause:** The speaker delivers a concrete noun and pauses for **600ms – 1000ms** (acoustic emphasis).

3. Leading-Edge Micro-Prefetch (Queue Depth = 1):

- Speculative prefetching fires *only* for the noun phrase currently crossing the speaker's lips. If the speaker changes direction without elevating the topic, the speculative cache entry is discarded within 3–4 seconds.
- If a sudden semantic shift occurs upon an intent marker, a Just-in-Time (JIT) query executes (<250ms latency).

2.3 Visual Retrieval, Filtering, and Studio Formatting

To ensure every fetched image looks like an intentional presentation slide:

- **Dual-Source Pipeline:**
- **Local Assets (`~/Assets`):** In-memory fuzzy path/metadata index (<50ms retrieval).

- **Curated Web Engine (Brave Image Search / Serper / Wikimedia Commons):**
Modifiers bias results toward presentation-ready assets:
- *Everyday Objects:* [Entity] + "studio product photography isolated white background OR transparent PNG"
- *Technical Systems:* [Entity] + "architecture diagram transparent OR schematic"
- **Local Proxy & Verification Guardrail (Rust):**
- Images are fetched asynchronously by Rust via `request` and stored in temporary local storage (`~/.cache/meetily/`).
- Bypasses third-party CORS, CORB, and hotlink protection.
- Rejects images failing quality filters: aspect ratios > 2.5:1 (banners), dimensions < 400px (icons/avatars), or invalid MIME types.

2.4 Control Scheme & Safety Ergonomics

- `Space` or `1 / 2 / 3` : Project the staged candidate to the Audience Stage.
- `Left Arrow` / `Right Arrow` : Step backward and forward through the visual reel.
- `B` or `Esc` (**Safety Blind / Panic Shutter**): Instantly triggers a soft-edge shutter wipe on the Stage into a neutral branded slate without interrupting screen-sharing.
- **Drop Anywhere** / `cmd + v` (**Scratchpad**): The entire Cockpit window serves as a drop target. Dragging a screenshot or pressing `cmd + v` instantly stages that asset into Slot 1 at 100% elevation.

3. Feature 2: Visually-Grounded Executive Summaries

3.1 The "Clipboard Test" Philosophy

A post-meeting summary must remain a portable, executive-grade document:

- **Strict Decoupling:** The generated text must never reference visuals directly ("*as seen in the chart above*" or "*refer to slide 4*" are forbidden).
- **Information Translation:** Visual context (from screen OCR or projected Stage slides) is translated into dense, explicit factual statements (e.g., translating "*this spike here*" into "*p99 read latency surged to 420ms*").
- If copied into an email, Jira ticket, or Slack channel, the plain text must be 100% understandable without images attached.

3.2 UI Structure: Split-Pane Inspector

- **Left Pane (Plain Markdown Text):** Distraction-free, scannable text summary with timestamp anchors (`[12:04]`). Includes a top-level `[Copy Summary]` button that extracts pure Markdown.

- **Right Pane (Synchronized Visual Context Reel):** An optional, collapsible side drawer that scrolls in lockstep with the text summary, displaying the single verified image anchor associated with that section.

3.3 The One-Anchor Selection Law

A summarized bullet condensing 5 minutes of speech and 4 slides must not be overloaded with images.

- **Hard Rule: 0 or 1 image per bullet point.** Default to 0.
- **Candidate Scoring Engine:** When a topic window contains multiple slides, the single winning anchor frame is selected by:
 1. **Dwell Time:** Slide with the longest active presentation duration (>45s).
 2. **Semantic Overlap:** High keyword/embedding match between the final generated bullet point and the slide's OCR text.
 3. **Density Threshold:** Slides with fewer than 4 distinct words (title/transition slides) or low visual complexity are disqualified.

3.4 Interleaved LLM Ingestion Contract

Visual context is fed into the local LLM (Ollama) as interleaved structural metadata:

```
[11:04:12] Speaker: "As you can see, this error threshold is completely breached."
Context on screen: { id: "SLIDE_4", window: "Datadog", text: "Service: auth-worker |

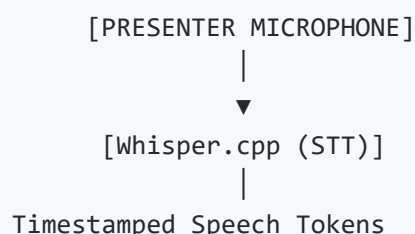
[11:04:30] Speaker: "We need to switch to this architecture pattern instead."
Context on screen: { id: "STAGE_REF_1", stage_projection: "Circuit Breaker Pattern"
```

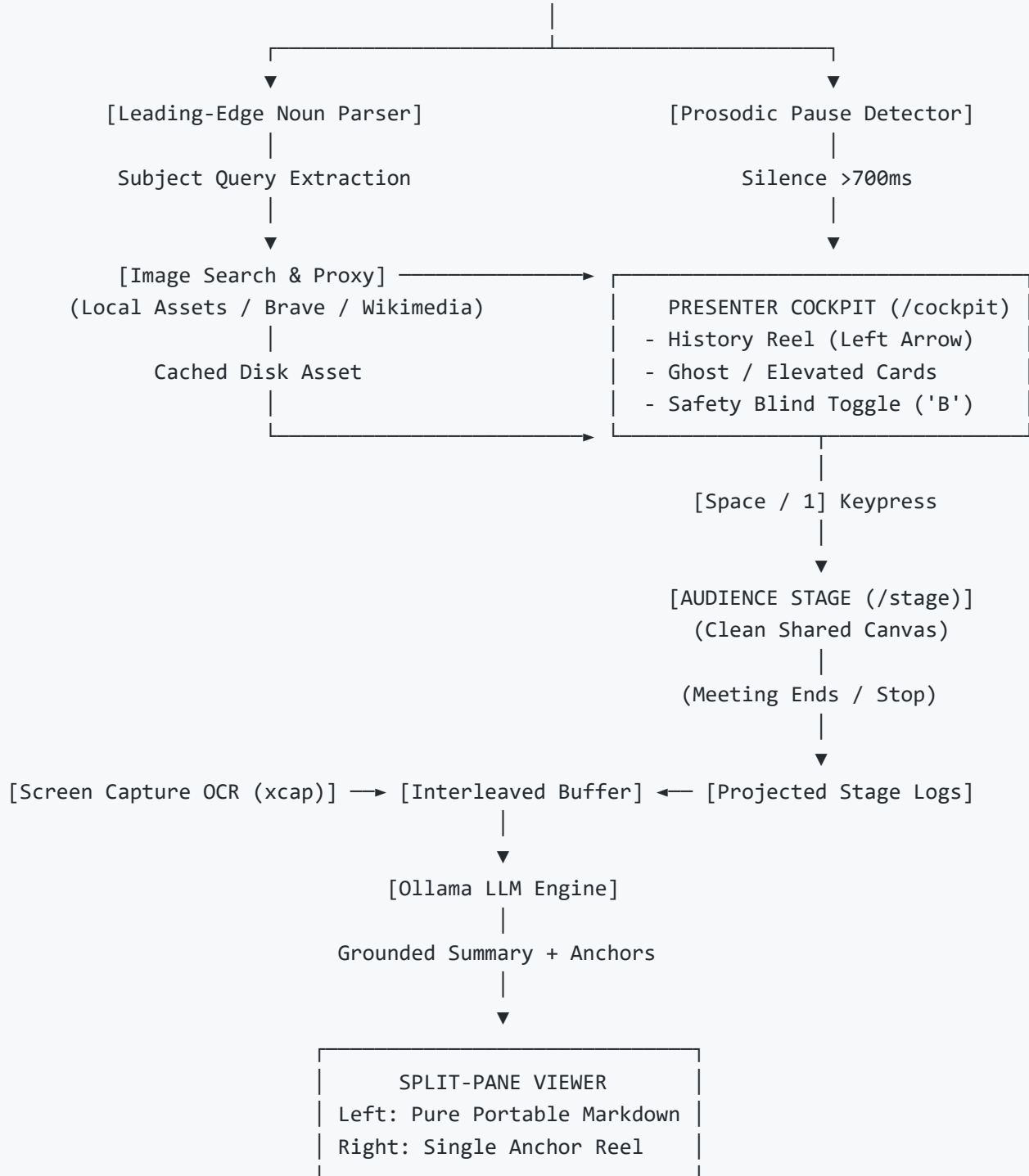
LLM System Directive:

"Resolve deictic references into factual assertions. Output clean markdown. For each bullet point, append at most one supporting visual anchor identifier using the syntax `<!-- anchor: SLIDE_ID -->` only when essential for technical verification, otherwise output `<!-- anchor: none -->`."

The frontend parses and strips these comments from the text buffer, rendering clean Markdown on the left while populating the side reel on the right.

4. End-to-End System Architecture





5. Technical Deliverable Summary

Component	Technology	Role
Shell & Core	Tauri v2 (<code>src-tauri</code>), Rust	Multi-window routing, IPC events, hotkey management, process isolation.
Capture & OCR	xcap , Apple Vision / tesseract-rs , image_hasher	Screen keyframe extraction, perceptual dHash deduplication, OCR parsing.
Audio & Speech	Whisper.cpp (Metal / CUDA bindings)	Low-latency local audio transcription, VAD pause monitoring.

Component	Technology	Role
Search & Proxy	request , tokio::fs	Local asset indexing, web search query formulation, download proxying, size/ratio validation.
Frontend UI	Next.js 14, Tailwind CSS, TypeScript	Cockpit HUD, borderless Stage projection window, Split-Pane summary inspector.
Inference	Ollama (llama3.1:8b or qwen2.5:7b)	Post-meeting multimodal plain-text synthesis and single-anchor frame mapping.